Supplementary Materials: A Unified Taxonomy of 19 AI System Types

HAMEEM MAHDI, Mayo Clinic, USA

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A Expert Validation Survey Instrument

The following survey was administered to 47 AI practitioners and researchers between June and September 2025 via a secure online platform. The study protocol was reviewed for ethical compliance. All responses were anonymised. Median completion time was 22 minutes.

A.1 Part 0: Informed Consent and Demographics

Q0.1 Do you consent to participate in this research study? Your responses will be anonymised and used solely for academic research.
☐ Yes ☐ No (exit survey)

Q0.2 What is your primary professional role?
☐ Academic researcher ☐ Industry practitioner ☐ Hybrid (both) ☐ Other: _____

Q0.3 How many years of experience do you have working with AI/ML systems?
☐ 0–2 ☐ 3–5 ☐ 6–10 ☐ 11–15 ☐ 16+

Q0.4 In which country are you primarily based? _____

Q0.5 Which AI system types do you work with most frequently? (Select all that apply)
☐ Generative ☐ Predictive/Discriminative ☐ Reinforcement Learning ☐ Conversational ☐ Recommendation/Retrieval ☐ Other: _____

A.2 Part 1: Taxonomy Completeness

Respondents were presented with the full list of 19 AI system types and their definitions.

Q1.1 Rate the overall completeness of the taxonomy (1 = Very Incomplete, 5 = Very Complete).
☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5

Q1.2 Are there any major AI system paradigms that are **missing** from the taxonomy?
☐ No, the taxonomy is sufficiently comprehensive
☐ Yes (please specify): _____

Q1.3 Are there any system types in the taxonomy that you consider **redundant** or that should be merged with another?
☐ No
☐ Yes (please specify which types and why): _____

Q1.4 Please rate the appropriateness of the six-cluster organisation (1 = Very Inappropriate, 5 = Very Appropriate).
☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5

Author's Contact Information: Hameem Mahdi, mahdi.hameem@mayo.edu, Mayo Clinic, Rochester, MN, USA.

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A.3 Part 2: Boundary Clarity

For each pair of system types listed below, rate how clearly their boundaries are defined (1 = Very Unclear, 5 = Very Clear).

Q2.1 Agentic vs. Autonomous (Non-Agentic):
☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5

Q2.2 Generative vs. Conversational:
☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5

Q2.3 Explainable (XAI) vs. Symbolic/Rule-Based:
☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5

Q2.4 Predictive/Discriminative vs. Analytical:
☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5

Q2.5 Cognitive/Neuro-Symbolic vs. Symbolic/Rule-Based:
☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5

Q2.6 Any additional comments on boundary clarity?
_____

A.4 Part 3: Practical Utility

Q3.1 How useful would this taxonomy be for **selecting** an appropriate AI approach for a new project? (1 = Not Useful, 5 = Very Useful)
☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5

Q3.2 How useful would this taxonomy be for **communicating** about AI systems with non-technical stakeholders? (1 = Not Useful, 5 = Very Useful)
☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5

Q3.3 How useful would this taxonomy be for **teaching** or curriculum development? (1 = Not Useful, 5 = Very Useful)
☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5

Q3.4 How useful would this taxonomy be for **policy-making** and regulation? (1 = Not Useful, 5 = Very Useful)
☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5

Q3.5 Open feedback: What improvements would make this taxonomy more useful in your work?
_____

B Literature Citation Analysis: Detailed Methodology

B.1 Database Search Strategy

Table 1 summarises the search queries used for each of the 19 system types across the four databases (Scopus, Web of Science, IEEE Xplore, ACM Digital Library).

B.2 Inclusion and Exclusion Criteria

Inclusion criteria:

- Peer-reviewed journal articles, conference papers, or workshop proceedings
- Published between January 2015 and December 2025
- Written in English
- Indexed in at least one of the four target databases

Exclusion criteria:

- Non-peer-reviewed preprints, blog posts, or technical reports

Table 1. Representative search queries for each AI system type (queries were adapted per database syntax)

System Type	Query Keywords (OR-combined)
Agentic	"agentic AI" OR "AI agent system" OR "goal-directed AI" OR "autonomous AI agent"
Autonomous (Non-Agentic)	"autonomous system" OR "self-governing AI" OR "autonomous control system" NOT "AI agent"
Reactive	"reactive AI" OR "stimulus-response system" OR "reactive agent"
Reinforcement Learning	"reinforcement learning" OR "RL agent" OR "reward-based learning" OR "policy gradient"
Symbolic / Rule-Based	"symbolic AI" OR "rule-based system" OR "expert system" OR "knowledge-based system"
Bayesian / Probabilistic	"Bayesian AI" OR "probabilistic model" OR "Bayesian network" OR "uncertainty quantification AI"
Cognitive / Neuro-Symbolic	"neuro-symbolic" OR "cognitive AI" OR "neural-symbolic" OR "hybrid reasoning"
Explainable (XAI)	"explainable AI" OR "XAI" OR "interpretable machine learning" OR "model interpretability"
Generative	"generative AI" OR "generative model" OR "GAN" OR "large language model" OR "diffusion model"
Predictive / Discriminative	"discriminative model" OR "classification system" OR "predictive AI" OR "supervised learning system"
Evolutionary / Genetic	"genetic algorithm" OR "evolutionary computation" OR "evolution strategy" OR "genetic programming"
Optimisation / OR	"AI optimization" OR "operations research AI" OR "constraint satisfaction" OR "mathematical programming"
Conversational	"conversational AI" OR "dialogue system" OR "chatbot" OR "virtual assistant"
Multimodal Perception	"multimodal AI" OR "multi-modal learning" OR "cross-modal" OR "vision-language model"
Physical / Embodied	"embodied AI" OR "robot learning" OR "physical AI" OR "sensorimotor AI"
Federated / Privacy-Preserving	"federated learning" OR "privacy-preserving AI" OR "differential privacy ML" OR "secure aggregation"
Recommendation / Retrieval	"recommendation system" OR "information retrieval AI" OR "collaborative filtering" OR "retrieval augmented"
Analytical	"AI analytics" OR "business intelligence AI" OR "data mining system" OR "analytical AI"
Scientific / Simulation	"scientific AI" OR "physics-informed neural network" OR "AI simulation" OR "computational science AI"

- Duplicate records (removed via DOI matching and title similarity with Levenshtein distance ≤ 0.05)
- Publications with fewer than 4 pages (to exclude abstracts and poster descriptions)
- Publications not primarily focused on AI system design or methodology

B.3 Deduplication Process

Records were first matched on DOI (exact match). Remaining records without DOIs were deduplicated using normalised title comparison (lowercase, punctuation-removed) with a Levenshtein distance threshold of 0.05. This

yielded a final corpus of 15,427 unique publications from an initial pool of 21,893 raw records (29.5% duplication rate across databases).

C Complete Case Study Mapping (25 Systems)

Table 2 presents the complete mapping of 25 production AI systems to taxonomy categories, including deployment scale and primary evidence source.

Table 2. Full case study mapping: 25 production AI systems across 8 industry sectors

#	System	Taxonomy Category	Sector	Scale	Source
1	AutoGPT	Agentic	Technology	Global	Open-source documentation
2	Devin (Cognition)	Agentic	Technology	Enterprise	Press coverage, demo
3	Waymo Driver	Autonomous (Non-Ag.)	Transportation	City-scale	Waymo safety reports
4	Tesla Autopilot	Autonomous (Non-Ag.)	Transportation	Global	SEC filings, technical blog
5	Siemens PLC	Reactive	Manufacturing	Global	Product documentation
6	AlphaGo / AlphaZero	Reinforcement Learning	Gaming/Research	Research	Nature publications
7	OpenAI Five	Reinforcement Learning	Gaming	Research	OpenAI blog
8	Google Knowledge Graph	Symbolic/Rule-Based	Technology	Global	Google AI blog
9	Palantir Foundry	Bayesian/Probabilistic	Defence/Finance	Enterprise	Palantir documentation
10	IBM Watson	Cognitive/Neuro-Symb.	Enterprise	Global	IBM research papers
11	SHAP / LIME	Explainable (XAI)	Finance/Health	Open-source	Original publications
12	Google Cloud XAI	Explainable (XAI)	Cloud	Enterprise	GCP documentation
13	GPT-4	Generative	Technology	Global	OpenAI technical report
14	Stable Diffusion	Generative	Creative	Global	Stability AI publications
15	PathAI	Predictive/Discrim.	Healthcare	Clinical	PathAI publications
16	Stripe Radar	Predictive/Discrim.	Finance	Global	Stripe engineering blog
17	Google NAS (NASNet)	Evolutionary/Genetic	Technology	Research	Google AI publications
18	Google OR-Tools	Optimisation/OR	Logistics	Open-source	Google documentation
19	ChatGPT / Claude	Conversational	Technology	Global	Technical reports
20	Google Gemini	Multimodal Perception	Technology	Global	Google DeepMind paper
21	da Vinci Xi	Physical/Embodied	Healthcare	Global	Intuitive Surgical reports
22	Boston Dynamics Spot	Physical/Embodied	Industrial	Enterprise	BD documentation
23	Google Gboard FL	Federated	Mobile	Billions	McMahan et al. 2017
24	Netflix Recommender	Recommendation	Entertainment	Global	Netflix tech blog
25	AlphaFold 2	Scientific/Simulation	Pharma/Research	Global	Nature publication

Table 3. Expert survey respondent demographics ($N = 47$)

Category	Group	Count
3*Role	Academic researcher	23
	Industry practitioner	19
	Hybrid	5
4*Experience (years)	0–5	8
	6–10	21
	11–15	12
	16+	6
5*Region	North America	16
	Europe	14
	Asia-Pacific	11
	Middle East	4
	Other	2

Table 4. Expert survey ratings summary (scale 1–5)

Dimension	Mean	SD	Median
Taxonomy completeness	4.2	0.7	4
Boundary clarity	3.8	0.9	4
Practical utility (selection)	4.1	0.6	4
Practical utility (communication)	4.3	0.5	4
Practical utility (teaching)	4.0	0.8	4
Practical utility (policy)	3.7	0.9	4
Cluster appropriateness	4.0	0.7	4

D Expert Survey: Detailed Results

D.1 Respondent Demographics

D.2 Quantitative Ratings

D.3 Top Qualitative Feedback Themes

(1) **Agentic vs. Autonomous boundary** (mentioned by 14/47 respondents): Respondents requested sharper criteria for distinguishing goal-directed agency from reactive autonomy.
(2) **XAI as overlay vs. standalone** (mentioned by 11/47): Several respondents noted XAI could be viewed as a meta-capability that applies across types rather than a standalone category.
(3) **Missing category: World-model-based AI** (mentioned by 9/47): Respondents suggested adding systems centred on internal world models (e.g., Sora, JEPA).
(4) **Temporal evolution** (mentioned by 7/47): Interest in how system types evolve and converge over time.
(5) **Positive reception of three-dimensional model** (mentioned by 31/47): The architecture–capability–deployment framework was highlighted as a particular strength.